FIN4080 Behavioral Finance
In-Class Demo Project

Lottery Stocks and the MAX Effect in China: A Guided Replication

The Chinese University of Hong Kong (Shenzhen)

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Data Source: CSMAR, Wind (full project); AKShare free interfaces (in-class demo subset).
Role: Guided replication — the instructor demonstrates the pipeline live in 15 minutes; students execute the full-scale version with the same code.

Project 58: Lottery Stocks and the MAX Effect in China: A Guided Replication

Goal.

This project replicates and extends the MAX effect of Bali, Cakici and Whitelaw (2011)—stocks with extreme recent daily returns subsequently underperform—in the full Chinese A-share cross-section, where retail investors dominate trading, price limits manufacture salient lottery events, and cumulative prospect theory predicts strong probability weighting. The instructor demonstrates the complete pipeline live in a 15-minute session on a cached teaching subset; students execute the full-scale version with the same code. Optional extensions (identification battery, dual-listed H-share mechanism test, cost-adjusted long-short backtest) are natural follow-ups for ambitious groups.

Presentation Questions.

- Does MAX robustly predict lower future returns in the full A-share universe, 2010–2024? How does the D10–D1 spread compare with the US benchmark of 0.7–0.9% per month?
- In a horse race among lottery measures—MAX, idiosyncratic skewness, and expected skewness—which one dominates?
- Is the effect concentrated among retail-held, high-turnover, low-priced stocks, and does it survive Fama–MacBeth controls for size, book-to-market, momentum, turnover, and idiosyncratic volatility?
- Is the premium stable across sub-periods (2015 crash, 2019 registration reform, post-2020 regime)?
- Do price-limit hit days drive the effect, and does MAX built on limit days price differently from MAX built on ordinary extreme days?

- Which interpretation fits better: behavioral probability weighting or rational skewness preference?

Suggested Analysis.

1. Universe: all A-shares, 2010–2024 (CSMAR/Wind); exclude ST/*ST stocks, the first six months after IPO, and suspended days; require a minimum number of valid trading days per month.
2. Construct lottery measures per stock-month: MAX (maximum daily return in month $t-1$); idiosyncratic skewness from rolling Fama–French three-factor residuals; expected skewness via the cross-sectional regression model of expected skewness; plus idiosyncratic volatility.
3. Univariate decile sorts by each measure (equal- and value-weighted); report D10–D1 spreads with Newey–West t-statistics; plot decile mean returns.
4. Fama–MacBeth regressions of month-t returns on each lottery measure with size, book-to-market, momentum, turnover, and IVOL controls; run the horse race with all three lottery measures together.
5. Bivariate sorts and interaction regressions: MAX × size, MAX × turnover, MAX × retail ownership (or small-trade share).
6. Sub-period robustness: pre/post the 2015 crash stabilization, pre/post the 2019 registration-system reform, and the post-2020 regime.
7. Price-limit decomposition: identify limit-hit days (+10%/+20% moves) and compare the pricing of MAX built on limit days versus ordinary extreme days.
8. Benchmark the Chinese spread against the US evidence and conclude with an explicit discussion of probability weighting versus rational skewness demand.

In-Class Demo Plan — 15 Minutes

0–2 min Motivate. Lottery tickets; A-share retail dominance (> 80% of volume); +10%/+20% price limits creating salient MAX events. State the question: do lottery-like stocks underperform?

2–4 min Data & setup. Show the 50-stock list and the CSV cache; explain why we cache (and how the full project swaps in CSMAR).

4–10 min Run the pipeline live. Execute the script section by section: load prices → build stock-month panel (MAX and next-month return) → decile sorts. Narrate each block in one sentence.

10–13 min Results. Show the two figures (decile bar chart; cumulative D10–D1 spread); read off the spread and the Newey–West t; compare with the US benchmark (0.7–0.9% per month).

13–15 min Debrief. One mechanism slide: probability weighting vs. rational skewness demand; preview the student extensions (horse race, price-limit decomposition, sub-periods).

Preparation: run `FIN4080_demo_max_effect.py` once before class to build the cache; copy `price_cache` to the classroom machine as a fallback.

Key Reference

Bali, T. G., Cakici, N., & Whitelaw, R. F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2), 427–446.